1.

Dropdown selection task

For any parameter w, an estimator $\hat{w}$ is consistent / efficient / unbiased if $\mathrm{E}(\hat{w}) = w$. The expected value / bias / mean squared error / variance of $\hat{w}$ can be decomposed as a sum of its expected value / bias / mean squared error / variance and its squared expected value / bias / mean squared error / variance.

Let $X_1, X_2, ..., X_n$ be random variables drawn independently from a distribution $p(\cdot; w)$. Consider $\frac{1}{n+1} \sum_{i=1}^{n} X_i$ as an estimator for the mean of the distribution $p(\cdot; w)$. This estimator is neither unbiased nor consistent / unbiased but not consistent / biased but consistent / both unbiased and consistent.

Now consider X_1 as an estimator for the mean of the distribution $p(\cdot; w)$. This estimator is neither unbiased nor consistent / unbiased but not consistent / biased but consistent / both unbiased and consistent.

Let $\{x_1, x_2, ..., x_n\}$ be a sample drawn independently from the distribution $p(\cdot; w)$. The function $\prod_{i=1}^{n} p(x_i; w)$ is called the probability / loss / density / likelihood function. Since $p(\cdot; w)$ is usually a probability / loss / density / likelihood function, the probability measure of the set $\{x_1, x_2, ..., x_n\}$ is in fact zero.

The Cramer-Rao lower bound relates the expected value / bias / mean squared error / variance of the estimator $\hat{w}$ to the inverse of its Fisher information. An estimator that reaches the Cramer-Rao lower bound is said to be consistent / efficient / unbiased.

2.

PCA:

You are given 4 data samples, each of dimension 3. The data matrix reads:

$$X = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -3 & 0 & 3 \\ -4 & 0 & 4 \end{pmatrix}$$

The eigenvectors of the covariance matrix of this data are:

$u_1 = \frac{1}{\sqrt{2}}(-1, 0, 1)^T$

$u_2 = \frac{1}{\sqrt{2}}(1, 0, 1)^T$

$u_3 = (0, 1, 0)^T$

These eigenvectors are already normalized and they are already correctly ordered corresponding to eigenvalues in descending order.

Compute the PCA projection (matrix) Y.

Enter the components of the projected third sample y_3 into the fields below.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

$y_3^{(1)} =$

$y_3^{(2)} =$

$y_3^{(3)} =$

3.

Assume you are given two binary random variables X and Y taking only the values 0 and 1 where their joint probability distribution is specified by the table below:

	Y = 0	Y = 1
X = 0	0.25	0.125
X = 1	0.25	0.375

This means, that e.g. $P(X = 0, Y = 0) = 0.25$, etc.

Compute the quantities (marginal probabilities, entropies, and mutual information) below. Use the **natural logarithm** in your calculations.

You can use a pocket calculator. Round the final values to two decimal places. (You may want to keep three-digit accuracy if you use intermediate results.) Number format example: 0.67

$P(X = 1) =$

$P(Y = 1) =$

$H(X) =$

$H(Y) =$

$I(X; Y) =$

4.

Which statements about Expectation Maximization (EM) and Maximum Entropy are correct?

Multiple answers are possible! Select one or more:

a. In the E-step, the hidden variables parameters are updated.

b. The entropy of a probability distribution is always larger equal 0 and smaller equal 1.

c. The Maximum Entropy approach requires minimal prior assumptions to assign a-priori-probabilities.

d. The EM algorithm can be applied to hidden Markov models and mixture models.

e. In the M-step, the model parameters are updated.

f. In EM, one deals with problems that have hidden variables as well as model parameters.

g. Hidden state models in general have analytical solutions for the likelihood.

h. The E-step increases the upper bound on the log-likelihood.

5.

Which statements about Independent Component Analysis (ICA) are correct?

Multiple answers are possible! Select one or more:

a. Independence can be quantified by the mutual information between components or the non-Gaussianity of components.

b. Fast ICA uses whitening and rotating the data.

c. Maximizing the negentropy minimizes the mutual information.

d. For non-Gaussian sources, the ICA solution is unique up to permutation and scaling.

e. Non-Gaussianity can be measured by the second and third cumulant.

f. For ICA, the number of sources must be smaller than, or equal to, the number of observations.

g. ICA components are ranked and orthogonal.

h. Infomax maximizes the mutual information between components.

6.

You are given two probability distributions p and q:

$$p(x = 0) = 0.8$$
$$p(x = 1) = 0.2$$

$$q(x = 0) = 0.5$$
$$q(x = 1) = 0.5$$

Compute the KL divergence $D_{\mathrm{KL}}(p \parallel q)$.

As for "log", use the natural logarithm.

You can use a pocket calculator. Round the final result to two decimal places. (If you use intermediate values, you should keep them with three digits precision.) Number format example: 0.67

$D_{\mathrm{KL}}(p \parallel q) =$

7.

Assume you want to do k-means on the following 2-dimensional dataset:

$X = \{(1,0), (2,-1), (1,-2), (0,-1), (2,0), (2,1), (3,0), (3,1)\}$.

The initial cluster centers are $\mu = (0, -2)$ and $\nu = (4, 1)$.

Compute the coordinates of the new cluster centers $\mu^{\text{new}} = (\mu_1^{\text{new}}, \mu_2^{\text{new}})$ and $\nu^{\text{new}} = (\nu_1^{\text{new}}, \nu_2^{\text{new}})$ after one interation of k-means.

Hint: Make a sketch first. The problem can mainly be solved graphically.

Enter the values with one decimal place.

$\mu_1^{\text{new}} =$ $\mu_2^{\text{new}} =$

$v_1^{\text{new}} =$ $v_2^{\text{new}} =$

8.

Which statements about Scaling and Projection Methods are correct?

Multiple answers are possible! Select one or more:

a. The objective of scaling and projection methods is to project data into a lower-dimensional space, e.g., to allow better model selection.

b. Projection pursuit performs projections such that the extracted signal is as Gaussian as possible.

c. Isomap uses a geodesic distance matrix, whose largest eigenvectors are the coordinates of the projected space.

d. The objective in t-SNE is minimization of the KL divergence between the similarities of the (high-dimensional) inputs and the (low-dimensional) mapped outputs.

e. Generative Topographic Mapping (GTM) is a non-linear latent variable model which maps latent variables to observations.

f. In locally linear embedding (LLE), each data point is described as a linear combination of its neighbors.

g. In self-organizing maps the objective is to minimize an energy (error) function.

h. Multidimensional Scaling (MDS) projects data to a higher-dimensional space in order to increase the distance between data points.

9.

Mixture model:

Assume the following case:

The prior probability of component j is 0.1.

The posterior probability of component j is 0.3.

The probability for observing a data sample x given the total parameter set θ is 0.2.

Compute the likelihood for observing a data sample x given component j and corresponding parameter vector.

You can use a pocket calculator. State your final result with two decimal places. Number format example: 0.67

Likelihood =

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Hierarchical Clustering

You are given the set of data points: $x = \{a, c, e, h, i, m, n, s\}$,

where the similarity between two points $s(x, x')$ is defined as the distance between the ordered letters in the alphabet,
i.e. $s(a, b) = 1$, $s(a, c) = 2$ and so on.

Perform agglomerative clustering of the data using average linkage -
the distance between clusters A and B is defined as $d_{\text{avg}}(A, B) = \frac{1}{n_A} \frac{1}{n_B} \sum_{a \in A} \sum_{b \in B} s(a, b)$.

Stop when the distance between clusters is greater than 7.

Answer the following questions (enter integer values):

How many clusters are formed in total?

What is the maximum number of data points in any one cluster?

What is the minimum distance between any two of the formed clusters?